UDAY SONAWANE

+1 (857) 421-9216 | udayds12mld@gmail.com | Boston, MA
linkedin.com/in/uday-sonawane | github.com/Udaydsmls | udaydsmls.github.io

EDUCATION

Northeastern University, Khoury College of Computer Sciences — Boston, MA
Master of Science, Computer Science | GPA 4.0/4.0 — Sept 2024 – May 2026
Coursework: Database Management Systems, Cloud Computing, Algorithms, Programming Design Paradigm, Machine Learning

Vellore Institute of Technology — Vellore, India
Bachelor of Technology, Computer Science | GPA 3.65/4.0 — Aug 2019 – Apr 2023

EXPERIENCE

Siemens Digital Industries Software — Pune, India
Software Engineer Intern — June 2023 – Jan 2024

- Built a Python ETL pipeline that ingests large system log files, parses and normalises them, classifies records into failure categories and ranks them by frequency. Cut manual investigation time by roughly 60%.
- Debugged production C++ and React services across distributed components on Linux, triaging system and console log failures and shipping fixes without breaking dependent teams' code paths.
- Traced a null value in a React log back through another team's code to the file producing it, then handed that team a reproducible issue they closed within days.
- Was the platform team's point of contact for three adjacent product teams on cross-cutting issues: understood what each team needed, scoped the trade-offs openly, agreed one approach, and delivered it.

Northeastern University, Khoury College of Computer Sciences — Boston, MA
Graduate Teaching Assistant, CS 5200 Database Management Systems — Jan 2025 – May 2026

- Graded assignments and projects for 60+ students per cohort across three semesters, always finishing ahead of the deadline, and wrote the grading rubrics the whole TA team marked against.
- Onboarded each term's new teaching assistants on rubric grading and on writing feedback a student can act on, and ran office hours explaining concepts to people meeting them for the first time.

PROJECTS

Archon, Retrieval over Technical Content | Python, FastAPI, LangGraph, Chroma, Pinecone, Weaviate, DynamoDB, Docker, GitHub Actions | GitHub

- Pulls relevant passages out of unstructured technical documents: sources ingested from text, PDF or URL, chunked into overlapping windows with source metadata, and retrieved by BM25 keyword search fused with dense vector search, since embeddings alone miss exact error strings and terms.
- Designed the API for operability: REST with JWT auth and validated schemas, streaming responses, per-run token, cost and latency tracking, and storage selected by environment variable so the same service runs locally or in the cloud.
- Quality is tested, not assumed: an evaluation suite scores answers against a golden dataset in CI and fails the build on regression.

Lakehouse Pipeline, Structured Data into Insight | Python, SQL, PostgreSQL, Kafka, Spark, dbt, Airflow | GitHub

- Turned raw events into analytics-ready tables: staging models standardise types and nulls, marts model users, products and orders dimensionally, and window functions compute retention, conversion and segmentation, gated by data-quality tests that halt the run.
- Exposed the insight through a FastAPI read API with validated parameters and documented every model and column, so a consumer gets an answer from a documented endpoint rather than raw tables.

Pulse, Services on Kubernetes | Go, TypeScript, Kafka, PostgreSQL, Docker, Kubernetes | GitHub

- Five services behind one gateway with clean API contracts, coordinating through Kafka with compensating rollback and idempotent handling, unit-tested per service in CI and deployed on Kubernetes.
- Kept data correct under concurrency by serialising contention in the database: reservation runs in one Postgres transaction with row-level locking, so two requests for the same record cannot both succeed.

OPEN SOURCE

God's Eye View | 49.3K GitHub stars, live spatial intelligence on a 3D globe | PR #664 merged | PR #875 in review

- Merged: fixed native dropdowns rendering near-white option text on a white popup in the dark theme across five menus, scoping the fix to the controls rather than the root so it would not reskin scrollbars that other stylesheets tune.
- In review: a NASA GIBS surface-temperature layer playing back MODIS monthly means by year, with click-to-read that decodes the published tile pixel against NASA's colour map, plus a fix to ledger tests that broke on every new layer PR.

Witch Hat Atelier Spell Simulator | Svelte | PR #8 merged

- Built a shape palette with rotate and resize controls and an arrange-mode toggle for users who find freehand drawing difficult, and added a small-screen overflow scrollbar in response to review. A maintainer built on it before merge.

SKILLS

Languages: Python, Java, TypeScript, JavaScript, C++17, SQL, Bash
Data and intelligence: Structured and unstructured extraction, log parsing and classification, RAG with hybrid retrieval, vector stores, SQL with window functions, PostgreSQL, dbt, Spark, Kafka, Airflow
Services and practice: REST and gRPC API design, FastAPI, Docker, Kubernetes, AWS, GitHub Actions CI/CD, unit testing (pytest, GoogleTest), code review, design documentation, troubleshooting on Linux
Fundamentals: Data structures and algorithms, operating systems, distributed systems, performance and scale, DBMS

ACHIEVEMENTS

Research and contests: Co-author of a multi-model study evaluating 7 LLMs, including automated extraction from unstructured responses; Codeforces rating 1439, placed 502nd of 23,000 participants in Codeforces Round 944